

W8D4 - Pratica

GameShell: Il gioco è strutturato in più livelli, l'avanzamento di livello richiede il completamento di particolari missioni.

```

kali@kali: ~
Session Actions Edit View Help

| The command |
| $ gsh help |
| displays the list of available (gsh) commands. |
+-----+
[mission 1] $ cd Castle
[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower
[use 'gsh help' to get a list of available commands]
[mission 1] $ cd First_floor
[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor
[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower
[use 'gsh help' to get a list of available commands]
[mission 1] $ gsh check

Congratulations, mission 1 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

```

Missione 1

salire in cima alla torre del castello.

[mission 1] \$ cd
Main_tower/First_floor/Second_floor/Top_of_the_tower/gsh check

Apprendimento del comando cd (Change Directory) per la navigazione gerarchica nelle cartelle.

```

kali@kali: ~
Session Actions Edit View Help

| Use the command |
| $ gsh help |
| to get the list of "gsh" commands. |
+-----+
[mission 2] $ cd..
cd..: command not found

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd
[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Castle
[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Cellar
[use 'gsh help' to get a list of available commands]
[mission 2] $ gsh check

Congratulations, mission 2 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

```

Missione 2

andare nella cantina del castello

[mission 2] \$ cd -cd Castle-cd Cellar-gsh check

Comprensione dei percorsi relativi (..) per risalire la struttura del file system.

```
kali@kali: ~
Session Actions Edit View Help

[ progress was saved in /home/kali/gameshell-save.sh ]
You have been teleported to .../World/Castle/Cellar

|-----|
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
|-----|

[mission 3] $ ls
barrel_of_apples

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd Castle/Main_building/Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ gsh check

Congratulations, mission 3 has been successfully completed!
```

Missione 3 : tornare alla stanza del trono utilizzando solo due comandi.
[mission 3] \$ cd
[mission 3] \$ cd Castle/Main_building/Throne_room/
gsh check

Utilizzo di cd con percorsi assoluti e scorciatoie per ottimizzare gli spostamenti.

```
~/Castle/Main_building/Throne_room
[mission 4] $ cd

~
[mission 4] $ cd Forest

~/Forest
[mission 4] $ mkdir Hut

~/Forest
[mission 4] $ cd Hut

~/Forest/Hut
[mission 4] $ mkdir Chest

~/Forest/Hut
[mission 4] $ gsh check

Congratulations, mission 4 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|-----|
|-----|
```

Missione 4 : costruire una capanna nella foresta e una cassa nella capanna
[mission 4] \$ cd
~
[mission 4] \$ cd Forest/
~/Forest
[mission 4] \$ mkdir Hut
~/Forest
[mission 4] \$ mkdir Hut/Chest -
gsh check

```
~/Forest/Hut
[mission 5] $ cd

~
[mission 5] $ cd Castle/Cellar

~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples bat_1 bat_2 spider_1 spider_2 spider_3

~/Castle/Cellar
[mission 5] $ rm spider_*

~/Castle/Cellar
[mission 5] $ gsh check

Congratulations, mission 5 has been successfully completed!
```

Missione 5 : tornare alla cantina e eliminare tutti i ragni lasciando i pipistrelli da soli
[mission 5] \$ ls

spider_1 spider_2 spider_3
~/Castle/Cellar
[mission 5] \$ rm spider_*
gsh check

Rivelazione dei file nascosti (spesso contrassegnati dal prefisso .) mediante le flag del comando ls.

```

~/Castle/Cellar
[mission 6] $ cd

~/Garden
[mission 6] $ cd Garden

~/Garden
[mission 6] $ ls
coin_1 coin_2 coin_3 Flower_garden Maze Shed

~/Garden
[mission 6] $ mv coin* ~/Forest/Hut/Chest

~/Garden
[mission 6] $ gsh check

Congratulations, mission 6 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|
--+-----+
| Use the command |
|                 |

```

Missione 6 : collezionare tutte le monete nel giardino e metterle nella cassa
[mission 6] \$ cd ../../Garden/
~/Garden
[mission 6] \$ ls

Flower_garden Maze Shed
~/Garden
[mission 6] \$ mv coin_*
../Forest/Hut/Chest
gsh check

```

~/Garden
[mission 7] $ cd

~/Garden
[mission 7] $ cd Garden

~/Garden
[mission 7] $ ls -A
.38821_coin_1 .64723_coin_3 .64811_coin_2 Flower_garden Maze Shed

~/Garden
[mission 7] $ mv .coin* ~/Forest/Hut/Chest
mv: cannot stat '.coin*': No such file or directory

~/Garden
[mission 7] $ mv .*coin* ~/Forest/Hut/Chest

~/Garden
[mission 7] $ mv .*_coin* ~/Forest/Hut/Chest
mv: cannot stat '.*_coin*': No such file or directory

~/Garden
[mission 7] $ gsh check

Congratulations, mission 7 has been successfully completed!

```

Missione 7 : colleziona tutte le monete nel giardino di fronte e metterle nella cassa nella foresta.
[mission 7] \$ ls -A
.56033_coin_3 .56399_coin_1
.59789_coin_2 Flower_garden
Maze Shed
~/Garden
[mission 7] \$ mv .*coin*
../Forest/Hut/Chest/
gsh check

Gestione di nomi file complessi (spazi o caratteri speciali) tramite il quoting o il carattere di escape.

```
kali@kali: ~  
Session Actions Edit View Help  
[mission 8] $ cd  
~  
[mission 8] $ cd Castle/Cellar  
~/Castle/Cellar  
[mission 8] $ ls -A  
10274_spider_43 17673_spider_21 26057_spider_28 3400_spider_40  
11371_spider_35 17851_spider_30 26207_spider_19 4196_bat_1  
11803_bat_4 1806_spider_27 26483_spider_20 426_spider_34  
13484_spider_17 18343_spider_4 27432_spider_7 6333_spider_33  
13530_spider_44 19896_spider_23 28159_spider_5 7172_spider_50  
14036_spider_24 22024_spider_8 28370_spider_6 8140_spider_12  
14175_spider_3 23049_spider_25 28437_spider_47 8565_spider_32  
14354_spider_31 23794_spider_11 29583_spider_26 8681_spider_49  
14446_spider_1 24301_spider_16 30355_spider_10 9195_spider_48  
14854_spider_37 24359_spider_13 31394_bat_3 9209_spider_38  
15504_spider_15 24709_spider_42 31483_spider_41 9563_spider_22  
16776_bat_2 24827_spider_45 31869_spider_9 9611_spider_39  
17168_bat_5 25345_spider_36 32105_spider_18 9670_spider_29  
17528_spider_2 25847_spider_14 32139_spider_46 barrel_of_apples  
~/Castle/Cellar  
[mission 8] $ rm *spider*  
~/Castle/Cellar  
[mission 8] $ gsh check  
Congratulations, mission 8 has been successfully completed!
```

Missione 8: rimuovi tutti i ragni che sono sulle ragnatele nella cantina
[mission 8] \$ ls -A ../Castle/Cellar/
...
~/Garden
[mission 8] \$ rm *spider*
gsh check

Ottimizzazione del lavoro tramite il completamento automatico, riducendo drasticamente le inefficienze nella digitazione.

```
kali@kali: ~  
Session Actions Edit View Help  
| explicit dot at the start of the pattern. |  
+-----+  
~/Castle/Cellar  
[mission 9] $ ls -A  
.11441_spider_6 .17168_bat_5 .26687_spider_18 .4212_spider_27  
.11803_bat_4 .17232_spider_45 .26883_spider_5 .6550_spider_20  
.11986_spider_4 .18610_spider_34 .27022_spider_14 .7016_spider_44  
.12495_spider_15 .18618_spider_10 .27286_spider_8 .7622_bat_5  
.12582_spider_38 .18793_spider_37 .282_spider_43 .7945_spider_12  
.1290_spider_46 .19118_spider_41 .28601_spider_42 .8059_spider_2  
.13700_spider_32 .19190_spider_3 .29694_spider_29 .8078_spider_48  
.14526_spider_21 .19225_spider_50 .29785_bat_3 .8374_spider_11  
.14612_spider_17 .19663_spider_19 .30491_spider_30 .8472_bat_2  
.15186_spider_24 .20231_spider_23 31394_bat_3 .9318_spider_22  
.15316_spider_35 .2036_spider_33 .31855_spider_16 .9494_spider_9  
.15450_spider_40 .20574_spider_31 .32036_spider_39 .9886_spider_7  
.16179_spider_25 .21328_spider_13 .32272_spider_36 barrel_of_apples  
.16328_spider_28 .21347_spider_49 .328_spider_26  
16776_bat_2 .23733_bat_1 .3437_spider_47  
.16978_bat_4 .24584_spider_1 4196_bat_1  
~/Castle/Cellar  
[mission 9] $ rm .*spider*  
~/Castle/Cellar  
[mission 9] $ gsh check  
Congratulations, mission 9 has been successfully completed!
```

Missione 9 : eliminare i ragni nascosti
[mission 9] \$ ls -A
[mission 9] \$ rm .*spider*
gsh check

Consultazione di file di testo estesi mediante l'interfaccia di less, permettendo una lettura agevole.

```

~/Castle/Cellar
[mission 10] $ cd

~
[mission 10] $ cd Castle/Great_hall

~/Castle/Great_hall
[mission 10] $ ls
43937_suit_of_armour      6974_stag_head    standard_2    standard_4
59094_decorative_shield  standard_1        standard_3

~/Castle/Great_hall
[mission 10] $ cp standard* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 10] $ gsh check

Congratulations, mission 10 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]
|

```

Missione 10 : fare una copia dei 4 standardi
[mission 10] \$ cd Castle/Great_hall
[mission 10] \$ ls
[mission 10] \$ cp standard_* ~/Forest/Hut/Chest/
gsh check

Filtraggio di informazioni specifiche all'interno di grossi file di testo con grep

```

~/Castle/Great_hall
[mission 11] $ ls
1300_tapestry_08      31324_tapestry_01      65419_stag_head    standard_3
17424_tapestry_04     36431_tapestry_05      6799_tapestry_02    standard_4
20769_tapestry_03     38314_decorative_shield 9979_tapestry_10
25511_tapestry_09     44087_suit_of_armour    standard_1
30258_tapestry_06     53178_tapestry_07      standard_2

~/Castle/Great_hall
[mission 11] $ cp tapestry* ~/Forest/Hut/Chest
cp: cannot stat 'tapestry*': No such file or directory

~/Castle/Great_hall
[mission 11] $ cp *tapestry* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 11] $ gsh check

Congratulations, mission 11 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

```

Missione 11: fare la copia anche degli arazzi
[mission 11] \$ ls
~/Castle/Great_hall
[mission 11] \$ cp *tapestry* ~/Forest/Hut/Chest/
gsh check

Verifica approfondita del percorso di lavoro corrente tramite pwd

```

~
[mission 12] $ cd

~/Castle
[mission 12] $ ls -l
total 20
drwxrwxr-x 7 kali kali 4096 May 29 03:56 Castle/
drwxrwxr-x 3 kali kali 4096 May 29 04:23 Forest/
drwxrwxr-x 5 kali kali 4096 May 29 04:33 Garden/
drwxrwxr-x 3 kali kali 4096 May 29 03:56 Mountain/
drwxrwxr-x 2 kali kali 4096 May 29 03:56 Stall/

~
[mission 12] $ cd Castle/Main_tower/First_floor

~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_vqralkjt ~/Forest/Hut/Chest

~/Castle/Main_tower/First_floor
[mission 12] $ gsh check

Congratulations, mission 12 has been successfully completed!

```

Missione 12: fare una copia del dipinto più vecchio
~/Castle
[mission 12] \$ cd Main_tower/First_floor/
[mission 12] \$ ls -l
[mission 12]cp
painting_vqralkjt../../Forest/Hut/Chest
gsh check

Analisi dei privilegi di accesso, distinguendo tra file leggibili, scrivibili o eseguibili.


```
~/Castle/Main_tower/First_floor
[mission 13] $ cal 1907
```

January							February							March						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4	5					1	2						1	2
6	7	8	9	10	11	12	3	4	5	6	7	8	9	3	4	5	6	7	8	9
13	14	15	16	17	18	19	10	11	12	13	14	15	16	10	11	12	13	14	15	16
20	21	22	23	24	25	26	17	18	19	20	21	22	23	17	18	19	20	21	22	23
27	28	29	30	31			24	25	26	27	28		24	25	26	27	28	29	30	
													31							

April							May							June								
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa		
			1	2	3	4	5	6				1	2	3	4							1
7	8	9	10	11	12	13	5	6	7	8	9	10	11	2	3	4	5	6	7	8		
14	15	16	17	18	19	20	12	13	14	15	16	17	18	9	10	11	12	13	14	15		
21	22	23	24	25	26	27	19	20	21	22	23	24	25	16	17	18	19	20	21	22		
28	29	30					26	27	28	29	30	31	23	24	25	26	27	28	29			
													30									

July							August							September								
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa		
			1	2	3	4	5	6					1	2	3	1	2	3	4	5	6	7
7	8	9	10	11	12	13	4	5	6	7	8	9	10	8	9	10	11	12	13	14		
14	15	16	17	18	19	20	11	12	13	14	15	16	17	15	16	17	18	19	20	21		
21	22	23	24	25	26	27	18	19	20	21	22	23	24	22	23	24	25	26	27	28		
28	29	30	31				25	26	27	28	29	30	31	29	30							

Missione 13 : scoprire il giorno legato al 06-20-1907

```
[mission 13] $ cal 1907
```

```
~/Castle/Main_tower/First_floor
```

```
[mission 13] $ gsh check
```

What was the day of the week for the 06-20-1907?

1 : Monday

2 : Tuesday

3 : Wednesday

4 : Thursday

5 : Friday

6 : Saturday

7 : Sunday

Your answer: 4

October							November							December						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4	5					1	2	1	2	3	4	5	6	7
6	7	8	9	10	11	12	3	4	5	6	7	8	9	8	9	10	11	12	13	14
13	14	15	16	17	18	19	10	11	12	13	14	15	16	15	16	17	18	19	20	21
20	21	22	23	24	25	26	17	18	19	20	21	22	23	22	23	24	25	26	27	28
27	28	29	30	31			24	25	26	27	28	29	30	29	30	31				

```
~/Castle/Main_tower/First_floor
[mission 13] $ echo "Thursday" > answer
```

```
~/Castle/Main_tower/First_floor
[mission 13] $ gsh check
What was the day of the week for the 06-20-1907?
```

1 : Monday

2 : Tuesday

3 : Wednesday

4 : Thursday

5 : Friday

6 : Saturday

7 : Sunday

Your answer: 4

Congratulations, mission 13 has been successfully completed!

```
~/Castle/Main_tower/First_floor
[mission 14] $ alias la='ls -A'
```

```
~/Castle/Main_tower/First_floor
[mission 14] $ gsh check
```

Congratulations, mission 14 has been successfully completed!

[progress was saved in /home/kali/gameshell-save.sh]

Missione 14: creare un alias "la" ed eseguire il comando "ls -A" per elencare tutti i file con due lettere

```
[mission 14] $ alias la='ls -A'
```

```
gsh check
```

Missione 15 : creare un file registro.txt nella tua cassa e scrivere un messaggio

```
[mission 15] $ nano
```

```
Forest/Hut/Chest/journal.txt
```

```
gsh check
```

```
~/Castle/Main_tower/First_floor
[mission 15] $ nano ~/Forest/Hut/Chest/journal.txt
```

```
~/Castle/Main_tower/First_floor
[mission 15] $ gsh check
```

Congratulations, mission 15 has been successfully completed!

```
~/Castle/Main_tower/First_floor
[mission 16] $ alias journal='nano ~/Forest/Hut/Chest/journal.txt'

~/Castle/Main_tower/First_floor
[mission 16] $ gsh check

Congratulations, mission 16 has been successfully completed!
```

Missione 16 : creare un alias "storico" in modo da editare il tuo file storico ovunque

```
[mission 16] $ alias journal='nano  
~/Forest/Hut/Chest/journal.txt'  
gsh check
```

Missione 17 : entrare nella tana dei ragni ed eliminare solo la regina in meno di 20 secondi.

```
[mission 17] $ cd
Castle/Cellar/.Lair_of_the_spider_queen\
HyBxNfhPqctuiOwg mFNIHbRkCdKwlco/
```

~/Castle/Cellar/.Lair_of_the_spider_queen
HvBxNfhPgctuiOwg mFNIHbRkCdKwllco/

```
[mission 17] $ ls -A
[mission 17] $ rm
eWmgLGXvKzBBwzHQ_spider_queen_XHb
AhqMAwTDgwWxe
[mission 17] gsh check
```

```
~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNlHBRkCdKwllco
[mission 18] $ keyes
keyes 5
gsh check
gsh reset
^C

~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNlHBRkCdKwllco
[mission 18] $ keyes
^C

~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNlHBRkCdKwllco
[mission 18] $ keyes 5
[1] 95178

~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNlHBRkCdKwllco
[mission 18] $ gsh check

Congratulations, mission 18 has been successfully completed!
```

Missione 18 :

[mission 18] \$ xeyes
^C

[mission 18] \$ xeyes &
[1] 95178

```
[mission 18] gsh check
```

```
~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNLHBRkCdKwllco
[mission 19] $ flarigo & flarigo & flarigo & gsh check
[2] 99541
[3] 99542
[4] 99543
```

```
      *
          *
    *
Let's have a look:
  *
        *
            *
    *
        *
            *
    *
        *
            *
    *
        *
            *
    *
        *
            *
    *
        *
            *
    .:.:.
    :.:~
    '.:.'
    ':
```

```
.
- \(/ -
./) \
.
```

Missione 19: accendere almeno 3 fuochi d'artificio

[mission 19] \$ flarigo & flarigo & flarigo &
gsh check

```
Great, that looked good!
[2] Done flarigo
[3]- Done flarigo
[4]+ Done flarigo

Congratulations, mission 19 has been successfully completed!
```

```
You can stop the incantation with Control-C.  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
....  
..C  
  
      ^.^-----.,,-,  
    <----->  
   /         \_____\-/  
  |             . , ; , ,  
  ||          -||| |  
  :==||       |||=:  
 -=#$$6%$#=:-  
     |        |:!  
 ____.-.#6$0%#6#~,._____
```

That's not working, try a different combination ...

```
~/Castle/Cellar/.Lair_of_the_spider_queen HyBxNfhPqctuiOwg mFNLHBRKcdKwllco  
[mission 20] $ charmiglio qwer  
.(-.* .. (-.(.
```

[illegible]

Missione 20 : aiutare il pirotecnico
trovando 4 lettere per lanciare i fuochi
giusti

[mission 20] \$ charmiglio qwer

[mission 20] \$ charmiglio zxcv

```
[mission 20] $ gsh check zxcv
```



```

+-----+
| Mission goal                                     |
|                                                     |
| Find the gold coins in the maze hidden in the garden and move them to |
| your chest.                                         |
|                                                     |
| Useful commands                                   |
|                                                     |
| find CONDITION                                     |
|   Search for files satisfying the condition, starting from your current |
|   working directory.                               |
|                                                     |
|   There are many possible conditions. They can constrain the           |
|   file names, size, modification date, etc. For example -name "PATTERN"|
|   -iname "PATTERN"                                |
|   are both related to file names.                  |
|                                                     |
| man COMMAND                                       |
|   Display the manual of the command.                |
|                                                     |
| Important key bindings:                          |
|   q          quit                                |
|   Space      scroll down one page                 |
|   / STRING   search for the string                |
|   n          find the next occurrence of the       |
|              search string                        |
|                                                     |
| Remark                                           |
|                                                     |
| The description of the ``find`` command is deliberately kept short. You |
| will find more information in the manual. (You can for exemple search for|
| the string "-name" in the manual.)                 |
+-----+

~/Garden/Maze
[mission 23] $ find . -name "gold_coin" -exec mv {} $GSH_CHEST/ ;
find: missing argument to `-exec'

~/Garden/Maze
[mission 23] $ find . -iname "*gold_coin*" -exec mv {} $GSH_CHEST/ \;

~/Garden/Maze
[mission 23] $ gsh check

Congratulations, mission 23 has been successfully completed!

```

Missione 23 : Trovare le monete d'oro
 nel labirinto nascosto nel giardino e
 portarle nel forziere.
 [mission 23] \$ find . -name "gold_coin"
 mv {} \$gsh_chest/ \;
 gsh chest


```
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.
10) Add 3 measures of boomslang skin to the cauldron.
11) Crush a bicorn horn into a fine powder.
12) Add 1 measure of the crushed horn to the cauldron.
13) Heat for 20 seconds at a high temperature.
14) Wave your wand then let potion brew for 24 hours.
15) Add 1 additional scoop of lacewings to the cauldron.
16) Stir 3 times, counter-clockwise.
17) Split potion into multiple doses, if desired.
18) Add a pieces of the person you wish to become.
19) Wave your wand to complete the potion.
vvvvvvvvvvvvvvvv
Elixir of youth
^^^^^^^^^^^^^^^^

WARNING: **do not** drink if you are older than 30.
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.

~/Mountain/Cave
[mission 26] $ cat Book_of_potions/page_01 Book_of_potions/page_02
vvvvvvvvvvvvvvvvvvvvvvvv
Transformation potion
^^^^^^^^^^^^^^^^^^^^^^^^^^^^

1) Boil water in a cauldron.
2) Add 3 measures of fluxweed to the cauldron.
3) Add 2 bundles of knotgrass to the cauldron.
4) Stir 4 times, clockwise.
5) Wave your wand then let potion brew for 80 minutes.
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.
10) Add 3 measures of boomslang skin to the cauldron.
11) Crush a bicorn horn into a fine powder.
12) Add 1 measure of the crushed horn to the cauldron.
13) Heat for 20 seconds at a high temperature.
14) Wave your wand then let potion brew for 24 hours.
15) Add 1 additional scoop of lacewings to the cauldron.
16) Stir 3 times, counter-clockwise.
17) Split potion into multiple doses, if desired.
18) Add a pieces of the person you wish to become.
19) Wave your wand to complete the potion.

~/Mountain/Cave
[mission 26] $ gsh check

Congratulations, mission 26 has been successfully completed!
```

```

~/Mountain/Cave
[mission 27] $ cat Book_of_potions/page_03 Book_of_potions/page_04
vvvvvvvvvvvvvvvv
Elixir of youth
^^^^^^^^^^^^^^^^
WARNING: **do not** drink if you are older than 30.
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

~/Mountain/Cave
[mission 27] $ cat Book_of_potions/page_03 Book_of_potions/page_04 | tail -n +5
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

~/Mountain/Cave
[mission 27] $ gsh check

Congratulations, mission 27 has been successfully completed!

```

Missione 27 : Il vecchio eremita ora vuole che trovi i passaggi per l'Elixir of Youth (l'Elisir di Giovinezza).

```

[mission 27] $ cat Book_of_potions/page_03
Book_of_potions/page_04
[mission 27] $ cat Book_of_potions/page_03
Book_of_potions/page_04 | tail -n +5
gsh check

```



```

~/Mountain/Cave
[mission 28] $ grep -i "distilled water" Book_of_potions/page_*
Book_of_potions/page_13:Distilled water

~/Mountain/Cave
[mission 28] $ cat -n Book_of_potions/book_of_potions/page_13:Distilled water
cat: 'Book_of_potions/book_of_potions/page_13:Distilled': No such file or directory
cat: water: No such file or directory

~/Mountain/Cave
[mission 28] $ cat -n Book_of_potions/page_13
 1 vvvvvvvvvvvvvvvv
 2 Distilled water
 3 ^^^^^^^^^^^^^^^^^
 4 1) Boil water in a big pot.
 5 2) Condense the vapor in a fresh container.
 6 3) Add minerals for a better taste (optional).
 7 vvvvvvvvvvv
 8 King's ale
 9 ^^^^^^^^^
10 1) Go to the castle's cellar and grab a bottle.
11 2) Serve cold, in a pint.

~/Mountain/Cave
[mission 28] $

~/Mountain/Cave
[mission 28] $ gsh check
You have not used a combination of 'head' and 'tail'!

Sorry, mission 28 hasn't been completed.

[ progress was saved in /home/kali/gameshell-save.sh ]

+-----+
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
+-----+

~/Mountain/Cave
[mission 28] $ head -n 6 Book_of_potions/page_13 | tail -n 3
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).

~/Mountain/Cave
[mission 28] $ gsh check

Congratulations, mission 28 has been successfully completed!

```

```

~/Mountain/Cave
[mission 29] $ gsh check

Congratulations, mission 29 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

```

Missione 28: L'eremita ha sete e vuole la ricetta per l'acqua distillata ("distilled water").

```

[mission 28] $ grep -i "distilled water"
Book_of_potions/page_*
[mission 28] $ cat -n
Book_of_potions/page_13
[mission 28] head -n 6
Book_of_potions/page_13 | tail -n 3

```

Missione 29 : Un folletto dispettoso ha lanciato un incantesimo che sporca tutto di carbone e, come dice il suggerimento, questo incantesimo è un processo in esecuzione.

```

[mission 29] $ ps
[mission 29] $ kill [numero_PID]
[mission 29] $ clear
[mission 29] $ gsh chesck

```



```
ps
  PID TTY          TIME CMD
  2471 pts/0    00:00:00 zsh
  2564 pts/0    00:00:00 bash
  2705 pts/0    00:00:00 bash
  2749 pts/0    00:00:00 bash
  8062 pts/0    00:00:00 bash
 10938 pts/0    00:00:03 bash
  95178 pts/0    00:00:15 xeyes
 254831 pts/0    00:00:00 spell
 255856 pts/0    00:00:00 ps

~/Mountain/Cave
[mission 30] $
      *#@*
      6_**/~
      !$-#

      *#@*
      6_**/~
      !$-#

      *#@*
      6_**/~
      !$-#

      *#@*
      6_**/~
      !$-#

pkill -15 spell; pkill -15 spell; pkill -9 spell; gsh check
You'll need to do better than that to kill my spell!

      *#@*
      6_**/~
      !$-#

      *#@*
      6_**/~
      !$-#

Congratulations, mission 30 has been successfully completed!
```

mission 30]\$ kill -15 247076; kill -9 247076
mission 30]\$kill 248274; kill -9 248274; gsh check

```
~/Mountain/Cave
[mission 31] $ cd

~
[mission 31] $ cd Castle/Cellar/

~/Castle/Cellar
[mission 31] $ ls
10051_snowflake  24661_coal      41185_coal      55239_snowflake
10104_coal       24698_snowflake 41214_snowflake 55280_coal
10134_snowflake  24891_coal      41356_snowflake 55381_coal
1034_coal        25098_snowflake 41564_snowflake 55739_coal
10396_coal       25258_coal      4189_snowflake  55804_coal
10427_snowflake  25583_coal      41913_coal      55935_coal
10445_coal       25944_snowflake 4196_bat_1      55953_coal
10738_snowflake  26072_snowflake 42016_snowflake 56000_coal
1074_coal        26114_snowflake 42235_snowflake 56084_coal
10855_coal       26376_snowflake 42461_coal      56181_coal
10971_coal       26416_snowflake 42503_snowflake 56255_coal
11029_snowflake  2650_snowflake  42649_snowflake 56381_snowflake
11059_snowflake  27164_snowflake 42738_snowflake 56417_coal
11193_coal       27213_snowflake 43208_snowflake 56679_snowflake
11196_coal       27254_coal      43280_coal      56680_coal
1128_snowflake   27293_coal      43316_snowflake 56959_snowflake
```

Missione 31 : sei nella cantina dove il demone e la fata stanno facendo a gara di magia. rimuovere i sortilegi senza toccare i fiocchi di neve
[mission 31] \$ pstree -p
[mission 31] \$ kill 95300 95301 95302
[mission 31] \$ ls -a
[mission 31] \$ rm *coal*
gsh chest

```

[mission 31] $ ps
  PID TTY          TIME CMD
  2007 pts/0    00:00:00 zsh
  4533 pts/0    00:00:00 bash
  4868 pts/0    00:00:00 bash
  95277 pts/0    00:00:00 nice_fairy
  95279 pts/0    00:00:00 mischievous_imp
  95284 pts/0    00:00:00 spell
  95285 pts/0    00:00:00 spell
  95286 pts/0    00:00:00 spell
  95287 pts/0    00:00:00 tail
  95300 pts/0    00:00:00 spell
  95301 pts/0    00:00:00 spell
  95302 pts/0    00:00:00 spell
  95303 pts/0    00:00:00 tail
 117319 pts/0    00:00:00 sleep
 117347 pts/0    00:00:00 sleep
 117412 pts/0    00:00:00 sleep
 117427 pts/0    00:00:00 sleep
 117496 pts/0    00:00:00 sleep
 117507 pts/0    00:00:00 sleep
 117516 pts/0    00:00:00 ps

```

```

~/Castle/Cellar
[mission 31] $ pstree 95279
mischievous_imp--3*[spell--sleep]
                  |
                  tail

```

```

~/Castle/Cellar
[mission 31] $ pstree 95277
nice_fairy--3*[spell--sleep]
             |
             tail

```

```

~/Castle/Cellar
[mission 31] $ pstree -p 95277
nice_fairy(95277)---spell(95284)---sleep(130155)
                  |
                  |---spell(95285)---sleep(129959)
                  |
                  |---spell(95286)---sleep(130075)
                  |
                  tail(95287)

```

```

~/Castle/Cellar
[mission 31] $ pstree -p 95279
mischievous_imp(95279)---spell(95300)---sleep(134537)
                        |
                        |---spell(95301)---sleep(134377)
                        |
                        |---spell(95302)---sleep(134441)
                        |
                        tail(95303)

```

```

~/Castle/Cellar
[mission 31] $ kill 95300 95301 95302

```

```

~/Castle/Cellar
[mission 31] $ pstree -p 95277
nice_fairy(95277)---spell(95284)---sleep(141753)
                  |
                  |---spell(95285)---sleep(141665)
                  |
                  |---spell(95286)---sleep(141717)
                  |
                  tail(95287)

```

```

~/Castle/Cellar
[mission 31] $ kill 95300 95301 95302

```

```

~/Castle/Cellar
[mission 31] $ pstree -p 95277
nice_fairy(95277)---spell(95284)---sleep(141753)
                  |
                  |---spell(95285)---sleep(141665)
                  |
                  |---spell(95286)---sleep(141717)
                  |
                  tail(95287)

```

```

~/Castle/Cellar
[mission 31] $ pstree -p 95279
mischievous_imp(95279)---tail(95303)

```

```

~/Castle/Cellar
[mission 31] $ ls
0_coal      24732_snowflake  40170_snowflake  55111_snowflake
10038_snowflake  24835_coal      40186_coal      55117_coal
10051_snowflake  24866_coal      40316_snowflake  55224_coal
10104_coal      24891_coal      40359_coal      55239_snowflake
10134_snowflake  25030_snowflake  4038_coal       55280_coal
10214_coal      25043_coal      4043_snowflake  5533_coal
10280_snowflake  25059_snowflake  40554_snowflake  55381_coal
10295_snowflake  25098_snowflake  40799_coal      55570_snowflake
1034_coal      25216_snowflake  40802_snowflake  5558_snowflake
10380_coal      25258_coal      40817_snowflake  55720_coal

```

```

~/Castle/Cellar
[mission 31] $ rm *coal

```

```

~/Castle/Cellar
[mission 31] $ ls
10038_snowflake  24732_snowflake  39129_snowflake  54240_snowflake
10051_snowflake  25011_snowflake  39168_snowflake  54285_snowflake
10134_snowflake  25030_snowflake  39176_snowflake  54395_snowflake
10280_snowflake  25059_snowflake  39367_snowflake  54479_snowflake
10295_snowflake  25098_snowflake  39428_snowflake  54486_snowflake
10427_snowflake  25216_snowflake  39539_snowflake  54493_snowflake
10613_snowflake  25370_snowflake  39551_snowflake  54602_snowflake
10629_snowflake  25677_snowflake  39553_snowflake  54604_snowflake
10671_snowflake  25858_snowflake  3959_snowflake   54620_snowflake
10738_snowflake  25901_snowflake  39604_snowflake  54690_snowflake

```

```
~/Mountain/Cave
[mission 31] $ gsh goal
```

Mission goal

The imp is comparing his magic with a fairy. They met in the cellar, and imp is conjuring lumps of coal while the fairy is conjuring delicate snowflakes.

Remove the imp's spells and the coal that litters the cellar, but don't touch the snowflakes!

Remark

Do not kill the imp or the fairy.

Useful commands

pstree PID

Print the list of processes with their parent / child relationship.

If no PID is given, show the list of all processes with their parent / child relationship.

Useful options:

- p show the PID of processes
- \$\$ This variable contains the PID of the shell and can be given as the PID.

kill N

Send the termination signal to process number N.

Remark: N is called PID, or "process identifier".

```
~/Mountain/Cave
```

```
[mission 31] $ pstree
```

```
systemd--ModemManager--3*[{ModemManager}]
          --NetworkManager--3*[{NetworkManager}]
          --3*[VBoxClient--VBoxClient--3*[{VBoxClient}]]
          --VBoxClient--VBoxClient--4*[{VBoxClient}]
          --VBoxDRMClient--5*[{VBoxDRMClient}]
          --VBoxService--8*[{VBoxService}]
          --accounts-daemon--3*[{accounts-daemon}]
          --agetty
          --colord--3*[{colord}]
          --cron
          --dbus-daemon
          --haveged
          --init -- /sbin/init
```

```
~/Castle/Cellar
[mission 31] $ gsh check

Congratulations, mission 31 has been successfully completed!

[1]- Killed "$GSH_TMP/${gettext "nice_fairy"}" (wd: ~/M
ountain/Cave)
(wd now: ~/Castle/Cellar)
[2]+ Killed "$GSH_TMP/${gettext "mischievous_imp"}" (wd
: ~/Mountain/Cave)
(wd now: ~/Castle/Cellar)
[ progress was saved in /home/kali/gameshell-save.sh ]
```

```
~/Castle/Cellar
[mission 32] $ gsh goal

+-----+
| Mission goal |
+-----+
| To get better in the magical art, one needs to know mental math. |
| Get ready, because Merlin is about to test your precision with sums. |
| Run the command ``gsh check`` to start. |
+-----+

~/Castle/Cellar
[mission 32] $ gsh check
77 + 37 = ?? 114
17 + 90 = ?? 107
27 + 28 = ?? 55
86 + 33 = ?? 119
30 + 34 = ?? 64

Congratulations, mission 32 has been successfully completed!
```

Missione 32: migliorare nelle arti magiche facendo calcoli
 [mission 32] \$gsh chest